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```
clc; clear all; close all
```

```
% pg 186
```

```
% theta = [r; K; z0]
```

Pt 1

```
thresh = 1e-6; % nu
```

Pt 2

i is the numel of time j is numel of params

```
time = 1:10;  
i = numel(time);  
j = 3;
```

```
syms r K z0 t
```

```
theta = [r; K; z0];
```

```
z = K / (1 + (K/z0 - 1) * exp(r * t));
```

```
%dz/dr
```

```
dz_dr = simplify(diff(z, r));
```

```
%dz/dK
```

```
dz_dK = simplify(diff(z, K));
```

```
% dz/dz0
```

```
dz_dz0 = simplify(diff(z, z0));
```

```
derivs = [dz_dr; dz_dK; dz_dz0];
```

```
theta_names = {"r"; "K"; "z0"; "t"};
```

```
for ii = 1:i % time
```

```
    theta_now = [output_function(time(ii)); time(ii)];
```

```

    for jj = 1:j % theta
        theta_star_j = theta_now(jj);
        df_dth_j_eqn = derivs(jj, 1);
        df_dth_j = subs(df_dth_j_eqn, theta_names, theta_now);
        S(ii, jj) = double(theta_star_j * df_dth_j);
    end
end

function out = output_function(time)
    r = sin(time);
    K = exp(time);
    z0 = 3;
    out = [r; K; z0];
end

```

Part 3

```

F = transpose(S) * S;
[F_eig_vect, F_eig] = eig(F);

[F_eig, I] = sort(diag(F_eig), 'descend');

F_eig_sorted(:, 1) = F_eig_vect(:, I(1));
F_eig_sorted(:, 2) = F_eig_vect(:, I(2));
F_eig_sorted(:, 3) = F_eig_vect(:, I(3));

```

Part 4

```

lam_1 = F_eig(1);
lam_p = F_eig(end);

m = [];
L = [];

if (lam_p/lam_1 > thresh)

    disp("All vars are identifiable");
else
    for i = 1:numel(F_eig) - 1
        % find m
        if ((F_eig(i)/lam_1) > thresh) && ((F_eig(i + 1)/lam_1) <=thresh)
            m = i;
        end

        % eigenvector corresponding with p
        vp = F_eig_sorted(:, end);
        [~,j] = max(vp);
        L = [L;j];
    end

    % This is part b. The algorithm is unclear on what needs to be
    % repeated, but this is my interpretation

```

```
% now, repeat for other indices

for t = 1:(m+1)
% eigenvector corresponding with cur index
    vp = F_eig_sorted(:, t);
    [~,j] = max(vp);
    L = [L;j];
end

L = unique(sort(L));

disp("The following indices are unidentifiable: " + string(L))

end

All vars are identifiable
```

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